Excel & Google Sheets Cheat Sheet (Modules 2–3)

Everything below works in Google Sheets and Excel 365. Examples use the ShopKart `orders_clean` sheet: P city_clean, Q unit_price_num, R quantity_clean, S discount_num, T payment_clean, U returned_clean, V revenue, W category. Data rows 2–5001.

Shortcuts

Action	Windows	Mac
Jump to end of data	Ctrl + Arrow	Cmd + Arrow
Select to end of data	Ctrl + Shift + Arrow	Cmd + Shift + Arrow
Undo / redo	Ctrl + Z / Ctrl + Y	Cmd + Z / Cmd + Shift + Z
Paste values only	Ctrl + Shift + V	Cmd + Shift + V
Toggle $ in a reference	F4 (Excel), F4 or Fn + F4 (Sheets)	Cmd + T (Excel), Fn + F4 (Sheets)
Copy formula down	Double-click the fill handle	same

Menus: View → Freeze → 1 row · Data → Create a filter · Data → Data cleanup → Remove duplicates (Excel: Data → Remove Duplicates) · File → Settings → Locale: India (₹ and lakh grouping).

References

Write	Copy down	Copy right
`A2`	row changes	column changes
`$A$2`	fixed	fixed
`A$2`	row fixed	column changes
`$A2`	row changes	column fixed

Cleaning

Problem	Formula
Extra spaces	`=TRIM(H2)`
Mixed capitals	`=PROPER(H2)` · `=UPPER(H2)` · `=LOWER(H2)`
Old names → standard	`=IFERROR(VLOOKUP(O2,city_map!$A$2:$B$6,2,FALSE),O2)`
Text price "₹1,299"	`=IF(ISNUMBER(F2),F2,VALUE(SUBSTITUTE(SUBSTITUTE(F2,"₹",""),",","")))`
Blank means zero	`=IF(G2="",0,G2)`
Blank → label	`=IF(I2="","Unknown",I2)`
Yes/No → Y/N	`=IF(L2="Yes","Y",IF(L2="No","N",L2))`
Known typo	`=IF(E2=99,1,E2)`
Flag duplicates	`=IF(COUNTIF($A$2:A2,A2)>1,"DUPLICATE","")`

Problem	Formula
DD-MM-YYYY text → date	`=DATE(VALUE(RIGHT(B2,4)),VALUE(MID(B2,4,2)),VALUE(LEFT(B2,2)))`
YYYY-MM-DD text → date	`=DATE(VALUE(LEFT(B2,4)),VALUE(MID(B2,6,2)),VALUE(RIGHT(B2,2)))`
Month number	`=MONTH(M2)`

Rules: never edit raw data (duplicate the tab); one helper column per problem; check each column with COUNTIF / filter; a blank rating stays blank.

Everyday formulas

Formula	Gives (ShopKart)
`=ROUND(R2*Q2*(1-S2),2)`	revenue of one order (O10001 = 1,737.60)
`=SUM(V2:V5001)`	total revenue ₹1,47,04,649.70
`=COUNTA(A2:A5001)`	5,000 orders (counts non-empty cells)
`=COUNT(Q2:Q5001)`	counts numbers only
`=AVERAGE(V2:V5001)`	₹2,940.93
`=MIN(...)` / `=MAX(...)`	₹139.30 / ₹1,49,990
`=COUNTIF(U2:U5001,"Y")`	687 returns
`=IF(V2>=5000,"Big","Normal")`	label per row (560 Big)

Conditional totals

```
=SUMIFS(sum_range, crit_range1, crit1, [crit_range2, crit2], ...)
=COUNTIFS(crit_range1, crit1, ...)
=AVERAGEIFS(avg_range, crit_range1, crit1, ...)
```

Question	Formula	Answer
Revenue in Delhi	`=SUMIFS($V$2:$V$5001,$P$2:$P$5001,"Delhi")`	₹23,23,405.40
Returns in Pune	`=COUNTIFS($P$2:$P$5001,"Pune",$U$2:$U$5001,"Y")`	146 of 451 (32.4%)
Avg Fashion order	`=AVERAGEIFS($V$2:$V$5001,$W$2:$W$5001,"Fashion")`	₹2,353.96
Orders ≥ ₹5,000	`=COUNTIFS($V$2:$V$5001,">=5000")`	560
Criteria from a cell	`=COUNTIFS($V$2:$V$5001,">="&E4,$V$2:$V$5001,"<"&F4)`	histogram bin

Tip: pin data ranges with $, leave the criteria cell relative, fill down, and add a total check row.

Lookups

Formula	Notes
`=XLOOKUP(D2,products!$A$2:$A$61,products!$C$2:$C$61,"Not`	Excel 2021/365 and Google Sheets

Formula	Notes
found")	
=VLOOKUP(D2,products!$A$2:$C$61,3,FALSE)	lookup column must be first; FALSE = exact
=INDEX(products!$C$2:$C$61,MATCH(D2,products!$A$2:$A$61,0))	works in every version

Pivot tables

Sheets: Insert → Pivot table → range → New sheet. Excel: Insert → PivotTable. Rows = one row per value (city_clean) · Columns = one column per value (category) · Values = SUM / COUNTA / AVERAGE (revenue) · Filters = keep some rows. "Show as % of row / column / grand total" for shares and rates.

Charts

Question	Chart
Compare categories	Bar / column (sorted)
Change over time	Line
Share of a whole (2–4 parts)	Pie (rarely)
Two numbers together	Scatter
Distribution of one number	Histogram

Title = the finding ("Pune returns 1 in 3 orders"), not just the topic.

Statistics functions (Module 3)

Measure	Formula	ShopKart example
Mean	=AVERAGE(range)	revenue ₹2,940.93
Median	=MEDIAN(range)	revenue ₹1,524.80
Mode	=MODE(range)	quantity 1
Mean with a condition	=AVERAGEIF(range,"<100000")	₹2,858.65 without bulk orders
Range	=MAX(r)-MIN(r)	delivery 10 days
Variance (sample)	=VAR(range) (also VAR.S)	delivery 2.34
Standard deviation	=STDEV(range) (also STDEV.S)	delivery 1.53 days
Quartiles	=QUARTILE(range,1) / 2 / 3	delivery 3 / 4 / 5
IQR	=QUARTILE(r,3)-QUARTILE(r,1)	delivery 2
Outlier fence	=Q3+1.5*IQR	delivery 8 → 36 orders above
Skewness	=SKEW(range)	revenue 10.5, delivery 0.63
Correlation	=CORREL(range1,range2)	discount–quantity +0.27; delivery–rating −0.33
Normal probability	=NORMSDIST(z)	used for the A/B p-value

Reading guide: mean ≫ median → right skew → report the median. |r| < 0.1 none, 0.1–0.3 weak, 0.3–0.6 moderate, > 0.6 strong. Correlation is not causation. p < 0.05 → difference probably real.

Common errors

Error	Usual cause
`#N/A`	lookup value not found (spelling, extra space) → wrap in IFERROR or clean with TRIM
`#VALUE!`	maths on text (e.g. "₹1,299") → convert with VALUE/SUBSTITUTE
`#DIV/0!`	dividing by zero or an empty cell
`#NAME?`	misspelt function name, or XLOOKUP in old Excel
`#REF!`	a referenced cell or sheet was deleted
Wrong totals after fill-down	missing $ on a range